Six-bar linkage by hand: workflow, formulas and traps

A condensed companion to the worked solution. The numbers in brackets are the worked example (OA = 180 mm, β = 20°, θ_1 = 10°, n_2 = 440 rpm), so you can check your method step by step.

0. Kit and page set-up

Ruler (mm), protractor, compass, two set squares, calculator in **degree** mode, and A4/A3 sheets. Work in mm on paper and in m in calculations. Write the scale on **every** drawing. Lay the space diagram out with O on the left, about 30 mm in from the edge and 30–40 mm above the bottom.

1. Data selection (about 5 min)

1. $\omega_2 = 2\pi n_2/60$ [46.0767 rad/s].
2. $OE = CD = 4\,OA$ [720 mm], $H = OA/4$ [45 mm], then choose AB and BE, then $AC = \frac{3}{5}AB$ [396 mm].
3. Check $OA + AB < OE + BE$ and $OA + OE < AB + BE$. With OA the shortest link, Grashof ($s + l \leq p + q$) makes the crank rotate fully.
4. A good starting choice is **AB ≈ 3.5–3.8·OA** and **BE ≈ 2·OA** (here 3.67·OA and 2.00·OA). It satisfies both inequalities with margin and keeps the transmission angle above ~50°.

2. Position analysis (Section 1)

- **One position (steps a–f):** OA at θ_2 → OE at θ_1 → arcs AB (centre A) and BE (centre E) meet at B (take the intersection *above* E, as in the handout figure) → AC at β *below* AB → horizontal line H below O → arc CD (centre C) meets it at D.
- **16 positions:** θ_2 = 60°, 82.5°, 105°, … (+22.5° each, CCW). Number A_1…A_{16} and repeat the numbers on B, C and D. Join successive points to get S_A (circle), S_B (arc), S_C (oval coupler curve) and S_D (straight line).
- **Rocker limits B′, B″ (exact construction):** the rocker stops when OA and AB are in line. Draw circles centred at O with radii **AB + OA** (extended) and **AB − OA** (folded); where they cut the arc of B (radius BE about E) are B′ and B″. [θ_4 = 93.6° at θ_2 = 35.2°, and 153.7° at θ_2 = 216.4°.]
- **Slider limits D_L, D_R:** where V_D = 0. Tabulate S against θ_2 from your 16 positions and interpolate around the largest and smallest S. [D_R = 1,223.6 mm at 10.4°, D_L = 924.5 mm at 176.8°, stroke 299.1 mm.]

3. Velocity polygon (Section 2)

- $V_A = \omega_2 \cdot OA$ [8.294 m/s], ⊥OA, pointing the way the crank turns (at θ_2 + 90° for CCW).
- Pick a scale that makes *oa* 100–160 mm long [1 m/s = 20 mm].
- **Order:** b = (⊥AB through a) ∩ (⊥BE through o) → c = (⊥AC through a) ∩ (⊥BC through b) → d = (horizontal through o) ∩ (⊥CD through c) → g_3 = (⊥AG_3 through a) ∩ (⊥BG_3 through b). g_2, g_4, g_5 are the mid-points of oa, ob, cd.
- **Angular velocity and sense:** $\omega = V_{rel}/r$. Rotate the link vector (e.g. A→B) by 90° anticlockwise; if the relative velocity (e.g. V_B/A) points that way the link turns CCW, otherwise CW. [ω_3 = 8.22 CW, ω_4 = 12.60 CCW, ω_5 = 1.40 CW rad/s]

4. Acceleration polygon (Section 3)

1. **Tabulate every normal component first:** $a^n = V_{rel}^2/r$, directed **towards** the centre of the relative rotation.
   - Ground-pivoted links use the absolute velocity: $a_A^n = \omega_2^2\,OA$ [382.2 m/s²], $a_B^n = V_B^2/BE$.
   - Floating links use the **relative** velocity: $a_{B/A}^n = V_{B/A}^2/AB$, not V_B^2/AB.

2. $\alpha_2 = 0 \Rightarrow a_A^t = 0$, so a_A is along A→O.
3. Scale so that o′a′ ≈ 150 mm [1 mm = 2.5 m/s²]. Draw each normal component from the right point, then the ⊥ line through its tip; intersections give b′, c′, d′ and g_3′.
4. $\alpha = a^t/r$, with the sense from the direction of a^t (same rule as for ω). [α_3 = 433, α_4 = 1,059, α_5 = 248 rad/s², all CCW]
5. A component shorter than 1 mm (here $a_{D/C}^n$ = 1.41 m/s²) can be omitted from the drawing, but state its value.

5. Instantaneous centres (Section 4)

$C = n(n-1)/2 = 15$. **Known (7):** I_{12} = O, I_{23} = A, I_{34} = B, I_{14} = E, I_{35} = C, I_{56} = D, and I_{16} at ∞ (vertical).

Step	IC	Line 1	Line 2
1	I_{13}	O–A ($I_{12}I_{23}$)	E–B ($I_{14}I_{34}$)
2	I_{24}	O–E ($I_{12}I_{14}$)	A–B ($I_{23}I_{34}$)
3	I_{15}	I_{13}–C ($I_{13}I_{35}$)	vertical through D ($I_{16}I_{56}$)
4	I_{36}	C–D ($I_{35}I_{56}$)	vertical through I_{13} ($I_{13}I_{16}$)
5	I_{25}	A–C ($I_{23}I_{35}$)	O–I_{15} ($I_{12}I_{15}$)
6	I_{45}	B–C ($I_{34}I_{35}$)	E–I_{15} ($I_{14}I_{15}$)
7	I_{26}	vertical through O ($I_{12}I_{16}$)	A–I_{36} ($I_{23}I_{36}$)
8	I_{46}	vertical through E ($I_{14}I_{16}$)	B–I_{36} ($I_{34}I_{36}$)

Velocities: $V_A = \omega_2 r_{12A}$; $\omega_3 = V_A/r_{13A}$; $V_B = \omega_3 r_{13B}$; $V_C = \omega_3 r_{13C}$; and $V_D = \omega_2 \cdot \overline{O\,I_{26}}$. The I_{26} route is the easy one, because I_{15} usually lies off the page [about 5 m away here]. Measure short radii carefully: a 0.5 mm error on a 15 mm radius is already 3 %.

6. Analytical method (Section 5): formulas to know

$$\mathrm{Re} = l_1\cos\theta_1 - l_2\cos\theta_2, \quad \mathrm{Im} = l_1\sin\theta_1 - l_2\sin\theta_2, \quad C = \frac{\mathrm{Re}^2 + \mathrm{Im}^2 + l_4^2 - l_3^2}{2l_4}$$

$$t = \frac{-\mathrm{Im} \pm \sqrt{\mathrm{Re}^2 + \mathrm{Im}^2 - C^2}}{C - \mathrm{Re}}, \qquad \theta_4 = 2\tan^{-1} t \ \ (\text{take } 0 < \theta_4 < 180^\circ),$$

$$\theta_3 = 2\tan^{-1}\frac{\mathrm{Im} + l_4\sin\theta_4}{l_3 + \mathrm{Re} + l_4\cos\theta_4}$$

$$\theta_5 = \sin^{-1}\frac{-H - y_C}{l_5}, \qquad S = x_C + l_5\cos\theta_5, \qquad y_C = l_2\sin\theta_2 + AC\sin(\theta_3 - \beta)$$

$$\omega_3 = \frac{l_2\omega_2\sin(\theta_4 - \theta_2)}{l_3\sin(\theta_3 - \theta_4)}, \qquad \omega_4 = \frac{l_2\omega_2\sin(\theta_2 - \theta_3)}{l_4\sin(\theta_4 - \theta_3)},$$

$$\omega_5 = \frac{-V_{Cy}}{l_5\cos\theta_5}, \qquad V_D = V_{Cx} - l_5\omega_5\sin\theta_5$$

$$\alpha_3 = \frac{CD - AF}{AE - BD}, \qquad \alpha_4 = \frac{CE - BF}{AE - BD} \ \ (\text{constants A–F of Week 7}),$$

$$\alpha_5 = \frac{l_5\omega_5^2\sin\theta_5 - a_{Cy}}{l_5\cos\theta_5}, \qquad a_D = a_{Cx} - l_5\alpha_5\sin\theta_5 - l_5\omega_5^2\cos\theta_5$$

Always check: substitute θ_3 and θ_4 back and confirm $|B - A| = AB$. This one-line check catches the sign slips in the lecture example (report Appendix A). Use ± signs on ω and α (+ = CCW).

7. Dynamic forces (Section 6)

- 14 unknowns: $F_{12x}, F_{12y}, F_{32x}, F_{32y}, F_{43x}, F_{43y}, F_{53x}, F_{53y}, F_{14x}, F_{14y}, F_{65x}, F_{65y}, F_{16y}, T_{12}$.
- Per link: $\sum F_x = ma_{Gx}$, $\sum F_y = ma_{Gy}$, $\sum(R \times F) + T = I_G\alpha$, with $R \times F = R_xF_y - R_yF_x$, and **R measured from G to the joint**.
- Replace F_{23} by $-F_{32}$, F_{34} by $-F_{43}$, F_{35} by $-F_{53}$ and F_{56} by $-F_{65}$ *before* building the matrix; that is where all the minus signs in [A] come from.
- Slider: $F_{16x} = \mu' F_{16y}$, where $\mu' = -\mu\,\mathrm{sign}(V_D)\,\mathrm{sign}(F_{16y})$; solve once with μ′ = 0 to get the signs.
- Check $T_{12}\,\omega_2 = \sum(m\,\mathbf{a}_G \cdot \mathbf{V}_G + I\alpha\omega)$ [T_{12} = 166.01 N·m].

8. Shaking force and balancing (Section 8)

$\mathbf{F}_s = \mathbf{F}_{21} + \mathbf{F}_{41} + \mathbf{F}_{61} = -\sum m_i \mathbf{a}_{Gi}$ and $M_s = T_{21} + \mathbf{R}_E \times \mathbf{F}_{41} + \mathbf{R}_D \times \mathbf{F}_{61}$ [1,813 N, 85.4 N·m]. Strategy: (1) split link 5 into $m_5/2$ at C and $m_5/2$ at D; (2) give links 2 and 4 the Berkof–Lowen counterweights so that the centre of mass of links 2–4 stays still; (3) the remaining force $-(m_6 + m_5/2)\,a_D$ is horizontal and reciprocating, so it can only be reduced (least squares, or an opposed slider); (4) add a flywheel, because the counterweights increase the torque fluctuation.

9. Traps that cost marks

1. Missing scale, units or arrowheads on polygons; unlabeled vectors (the rubric asks for “scale, labelling and directions”).
2. Using the absolute instead of the relative velocity in a^n, or drawing a^n away from the centre.
3. Mixing mm and m (ω × mm gives mm/s).
4. Taking the wrong root for θ_4 (B below E), or using $+H$ for a slider below O.
5. Measuring kinematics at a WM body’s centre of mass instead of at a point placed on the joint, which gives ≈ 50 % low values (FAQ).
6. Leaving gravity on in WM when comparing with the matrix method (here ≤ 6.2 N·m difference in T_{12}).
7. Not stating the G_3 interpretation (“β/2 and AB/2”: bisector, AB/2 from A).